

AKSHAYKUMAR KATHIRVELU

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PROFESSIONAL SUMMARY

Analytical cybersecurity professional with experience across Security Operations, MSSP, and network security environments. Skilled in SIEM monitoring and security event correlation using Splunk and Microsoft Sentinel, with hands-on experience in alert triage, log analysis, threat detection, and analysis of suspicious network and authentication activity to identify potential threats, support incident response, and prioritize timely escalation. Proficient in translating technical findings and security risks into actionable recommendations aligned with enterprise security, compliance, and operational requirements.

TECHNICAL SKILLS

Security Operations: Splunk (SPL), Microsoft Sentinel (KQL), Wazuh, Wireshark, Nmap, Nessus, Qualys, MITRE ATT&CK, NIST Incident Response Lifecycle, Alert Triage, Log Analysis, IOC Analysis, Incident Investigation, Threat Detection

Cloud & IAM: AWS (GuardDuty, CloudTrail, Security Hub, IAM), Azure (Defender for Cloud, Defender for Endpoint, Entra ID), GCP (Security Command Center, Cloud Logging)

Networking: HPE Comware, Aruba AOS-CX & ProCurve, Cisco Routing and Switching; 802.1X, RADIUS, TACACS+, IPsec, GRE, SSL/TLS, VLANs, Subnetting, DHCP, OSPF, BGP

OS & Scripting: Windows, Linux (RHEL, Ubuntu, Kali), Python, PowerShell, Bash, SQL, Git

EXPERIENCE

Graduate Teaching Assistant, Information Security | *George Mason University, Fairfax, VA* **Aug2025–May2026**

- Mentored **200+ students** through hands-on labs in malware analysis, phishing simulation, and Metasploit exploitation, guiding payload generation, listener configuration, and Meterpreter sessions across weekly lab sessions and office hours.
- Assessed **60+ assignment sets per semester** spanning vulnerability case studies, cryptanalysis, risk assessments, and breach analysis reports, evaluating student work against NIST SP 800-aligned security assessment methodologies.

Cybersecurity SME Intern, MSSP | *Comtech LLC, Reston, VA* **May2025–Aug2025**

- Designed a managed security services package covering alert triage, endpoint security, network and web monitoring, and gap assessment methodologies, delivered across **5+ healthcare and government clients**.
- Established the organization's **first CMMC compliance framework**, mapping control requirements to service offerings and producing control-gap documentation for client engagements.
- Applied STRIDE and DREAD threat modeling across client security requirements and authored **three technical whitepapers** that supported a government and healthcare contract proposal.

Network Security Engineer, Tier 1 → Tier 2 | *Movate (HPE Aruba Networking), Chennai, India* **Sep2021–Jul2024**

- Investigated and resolved **90+ security incident tickets monthly** plus a daily priority escalation across infrastructure of **2,500+ switches and 3,000+ users**, using root-cause analysis, Wireshark and Nmap packet inspection, and controlled lab replication that surfaced misconfigurations including QoS and DHCP policy failures.
- Reduced MTTR by **80%**, from hours to minutes, by centralizing switch syslog and security event monitoring in Splunk and escalated complex cases to engineering with packet-level evidence.
- Completed **200+ vulnerability assessments** with Qualys and Nessus, prioritizing findings by severity and exploitability and coordinating remediation of **100+ vulnerabilities** with infrastructure teams.
- Applied 802.1X, RADIUS, and TACACS+ access controls and configured IPsec and GRE tunnels across HPE, ArubaOS-CX, and ProCurve environments, while mentoring **20+ Tier 1 engineers** on incident troubleshooting and network hardening.

PROJECTS

AWS CloudTrail Threat Detection, *CloudWatch* | *Sigma* | *MITRE ATT&CK*

- Built and validated three detections for root account usage, IAM privilege escalation, and log tampering as CloudWatch metric filters and portable Sigma rules with measured alert times of **84–208 seconds**; reconstructed the incident from logs alone, filtering **680 events to 11 state-changing actions**, containing it in 94 seconds and documenting 10 severity-ranked coverage gaps.

Wireshark Network Threat Investigation, *Kali Linux* | *Ubuntu* | *VMware*

- Analyzed packet-level traffic tied to reconnaissance, brute-force activity, and C2/DGA behavior, mapping **10+ MITRE ATT&CK techniques** and building a reusable detection-filter reference for repeatable investigations.

EDUCATION & CERTIFICATIONS

M.S. in Applied IT, Cybersecurity | *George Mason University, Fairfax, VA* | *GPA: 3.77* **Aug2024–May2026**

B.E. in Electronics and Communication | *JMEC – Anna University, Chennai, India* | *GPA: 3.08* **Aug2017–May2021**

Certifications: CompTIA Security+ | AWS Cloud Practitioner | AWS AI Practitioner | ISC2 CC | Microsoft Azure Fundamentals | Fortinet Associate in Cybersecurity | Fortinet Fundamentals in Cybersecurity